

AI-Powered Healthcare Data Insights and Dashboarding

The given excel data set contains 3 sheets, customer name, medical examination and hospitalisation details.

A	B	C	D
Customer ID	name		
Id1	Hawks, Ms. Kelly		
Id2	Lehner, Mr. Matthew D		
Id3	Lu, Mr. Phil		
Id4	Osborne, Ms. Kelsey		
Id5	Kadala, Ms. Kristyn		
Id6	Baker, Mr. Russell B.		
Id7	Macpherson, Mr. Scott		
Id8	Hallman, Mr. Stephen		
Id9	Moran, Mr. Patrick R.		
Id10	Benner, Ms. Brooke N.		
Id11	Fierro Vargas, Ms. Paola Andrea		
Id12	Franz, Mr. David		
Id13	Foster, Mr. Wade		
Id14	Tenorio, Mr. Franklin		
Id15	Rios, Ms. Leilani M.		
Id16	Viau-Dupuis, Mr. Philippe		
Id17	Cronin, Ms. Jennifer A.		
Id18	Noordstar, Ms. Christina M.		

A	B	C	D	E	F	G	H
Customer ID	BMI	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	smoker
Id1	47.41	7.47	No	No	No	No major surgery	yes
Id2	30.36	5.77	No	No	No	No major surgery	yes
Id3	34.485	11.87	yes	No	No		2 yes
Id4	38.095	6.05	No	No	No	No major surgery	yes
Id5	35.53	5.45	No	No	No	No major surgery	yes
Id6	32.8	6.59	No	No	No	No major surgery	yes
Id7	36.4	6.07	No	No	No	No major surgery	yes
Id8	36.96	7.93	No	No	No		3 yes
Id9	41.14	9.58	yes	No	Yes		1 yes
Id10	38.06	10.79	No	No	No	No major surgery	yes
Id11	37.7	5.96	yes	No	No		2 yes
Id12	42.13	11.9	No	No	No	No major surgery	yes
Id13	40.92	8.41	No	No	No	No major surgery	yes
Id14	40.565	7.02	No	No	No	No major surgery	yes
Id15	36.385	7.59	yes	No	No		2 yes

A	B	C	D	E	F	G	H	I
Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID
Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013
Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013
Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013
Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013
Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013
Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012
Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013
Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013
Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012
Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013
Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013
Id2324	1999	Dec	26	0	700	?	tier - 3	R1013
Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013
Id2322	2002	?	19	0	750	tier - 3	tier - 1	R1012
Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013
Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013

Data Cleaning:

Remove duplicate

In the first step of data cleaning, count of Customer ID is checked in all three data sheets. The customer names sheet and medical examination sheet have the same Customer Ids. But the hospitalisation details data set contains 8 extra Ids.

A	B	C	D	E	F	G	H	I	J
Customer	Imputed customer	year	month	date	children	charges	Hospital tier	City tier	State ID
?	?	2004	Nov		6	0	1137.01 tier - 3	tier - 1	R1013
6	?	1999	Jun		9	1	2775.19 tier - 2	tier - 1	R1012
3	?	1985	Dec		20	2	6203.9 tier - 1	tier - 2	R1012
15	id2444	1987	Nov		27	2	20984.09 tier - 2	tier - 2	R1015
10	id3444	2004	Nov		1	2	34303.17 tier - 1	tier - 3	R1013
11	?	2000	Oct		13	0	35585.58 tier - 1	tier - 2	R1011
10	?	1992	Oct		6	0	36837.47 tier - 1	tier - 2	R1011
14	?	1991	Nov		22	2	38711 tier - 1	tier - 3	R1011
15									

Then the unknown Customer Ids were checked with the “Charges” column. Except the Customer ID all the other data were same in 5 Ids. So we considered that 5 unknown Ids are duplicate. Using the “Remove duplicates” that 5 Ids are removed. But the other 3 Ids are included in the data set.

A	B	C	D	E	F	G	H
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State
Id2042	09-Jun-1999	23	1	\$2,775.19	tier - 2	tier - 1	R1012
Id2041	29-Aug-1999	23	1	\$2,775.19	tier - 2	tier - 1	R1012
Unknown 2	09-Jun-1999	23	1	\$2,775.19	tier - 2	tier - 1	R1012

A	B	C	D	E	F	G	H
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State
Id1605	20-Dec-1985	37	2	\$6,203.90	tier - 1	tier - 2	R1012
Id1604	22-Sep-1985	37	2	\$6,203.90	tier - 2	tier - 2	R1012
Unknown 3	20-Dec-1985	37	2	\$6,203.90	tier - 1	tier - 2	R1012

A	B	C	D	E	F	G	H
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State
Id209	13-Oct-2000	22	0	\$35,585.58	tier - 1	tier - 2	R1011
Id208	19-Nov-2000	22	0	\$35,585.58	tier - 2	tier - 3	R1011
Unknown 4	13-Oct-2000	22	0	\$35,585.58	tier - 1	tier - 2	R1011

A	B	C	D	E	F	G	H	I
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State	
Id180	06-Oct-1992	30	0	\$36,837.47	tier - 1	tier - 2	R1011	
Id179	04-Jul-1992	30	0	\$36,837.47	tier - 2	tier - 3	R1011	
Unknown 5	06-Oct-1992	30	0	\$36,837.47	tier - 1	tier - 2	R1011	

A	B	C	D	E	F	G	H	I
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State	
Id136	22-Nov-1991	31	2	\$38,711.00	tier - 1	tier - 3	R1011	
Id135	02-Nov-1991	31	2	\$38,711.00	tier - 2	tier - 2	R1011	
Unknown 6	22-Nov-1991	31	2	\$38,711.00	tier - 1	tier - 3	R1011	

Data imputation

The number of missing values marked with "?" in each column of the "Medical Examinations" table and "Hospitalization Details" tables are checked by clicking the "?" in the filter.

Customer Id: The missing customer Id is imputed as "Unknown".

A	B	C	D	E	F	G	H
Customer	Date of Birth	Age	children	charge	Hospital tier	City tier	State
Unknown	06-Nov-2004	18	0	1137.01	tier - 3	tier - 1	R1013

Smoker:

In medical examination sheet 2 smoker values are missing.

A	B	C	D	E	F	G	H	I	J
Customer	BMI	HBA1C	Heart Issue	Any Transplan	Cancer histo	NumberOfMajorSurgery	smoke		
Id560	23.98	4.9	No	No	No	No major surgery	?		
Id635	25.175	4.96	No	yes	No		1 ?		

Based on the given instructions the missing values in the 'smoker' was imputed by the most frequently occurring values in the respective columns. Using the functions IF, INDEX, MODE and MATCH the missing values are imputed.

Clipboard Font Alignment Number Formatting Table Styles Format Cells										
fx =IF(H2="?",INDEX(\$H\$2:\$H\$2336,MODE(MATCH(\$H\$2:\$H\$2336,\$H\$2:\$H\$2336,0))),H2)										
	B	C	D	E	F	G	H	I	J	K
	BMI	HBA1	Heart Issu	Any Transplan	Cancer histo	NumberOfMajorSurgerie	smoke	Smoker		
	47.41	7.47	No	No	No	No major surgery	yes	yes		
	30.36	5.77	No	No	No	No major surgery	yes	yes		
	34.485	11.87	yes	No	No		2 yes	yes		
	38.095	6.05	No	No	No	No major surgery	yes	yes		
	35.53	5.45	No	No	No	No major surgery	yes	yes		
	32.8	6.59	No	No	No	No major surgery	yes	yes		
	36.4	6.07	No	No	No	No major surgery	yes	yes		
	36.96	7.93	No	No	No		3 yes	yes		
	41.14	9.58	yes	No	Yes		1 yes	yes		
	38.06	10.79	No	No	No	No major surgery	yes	yes		
	37.7	5.96	yes	No	No		2 yes	yes		
	42.13	11.9	No	No	No	No major surgery	yes	yes		
	40.92	8.41	No	No	No	No major surgery	yes	yes		
	40.565	7.02	No	No	No	No major surgery	yes	yes		
	36.385	7.59	yes	No	No		2 yes	yes		

Month: In Hospitalization Details 3 month values are missing. Based on the given instructions the missing values of 'month' was filled with "September" using the formula =IF (C2= "?", "Sep", C2)

Clipboard Font Alignment Number Styl										
fx =IF(C7="?", "Sep", C7)										
	A	B	C	D	E	F	G	H	I	J
	Customer	year	month	month	date	childre	charge	Hospital tie	City tie	State I
	Id2322	2002	?	Sep	19	0	750	tier - 3	tier - 1	R1012
	Id2318	1996	?	Sep	18	0	770.38	tier - 3	?	R1012
	Id3	1970	?	Sep	11	3	60021.4	tier - 1	tier - 1	R1012

Year:

In Hospitalization Details, two year values are missing. Based the given instructions the missing year values are replaced with its average rounded value using the formula =IF(B2= "?",ROUND(AVERAGE(\$B\$2:\$B\$2344),0),B2)

Clipboard Font Alignment Number Styles										
fx =IF(B1051="?",ROUND(AVERAGE(\$B\$2:\$B\$2344),0),B1051)										
	A	B	C	D	E	F	G	H	I	J
	Customer	year	year	month	date	childre	charge	Hospital tie	City tie	State I
	Id1289	?	1983	Jul	24	0	8534.67	tier - 2	tier - 3	R1024
	Id1286	?	1983	Dec	12	1	8547.69	tier - 2	tier - 1	R1013

Hospital tier and City tier:

The missing values in 'hospital tier', and 'city tier' are imputed by the most frequently occurring values in the respective columns by the given instructions. Using the functions IF, INDEX, MODE and MATCH the missing values are imputed.

=IF(G13="?",INDEX(\$G\$2:\$G\$2344,MODE(MATCH(\$G\$2:\$G\$2344,\$G\$2:\$G\$2344,0))),G13)									
Year	Month	date	children	charge	Hospital tier	Hospital tier	City tier	State	
1999	Dec	26	0	700	?	tier - 2	tier - 3	R1013	

=IF(G2="?",INDEX(\$G\$2:\$G\$2344,MODE(MATCH(\$G\$2:\$G\$2344,\$G\$2:\$G\$2344,0))),G2)									
Year	Month	date	children	charge	Hospital tier	Hospital tier	City tier	State	
1992	Jul	9	0	563.84	tier - 2	tier - 2	tier - 3	R1013	
1992	Nov	30	0	570.62	tier - 2	tier - 2	tier - 1	R1013	
1993	Jun	30	0	600	tier - 2	tier - 2	tier - 1	R1013	
1992	Sep	13	0	604.54	tier - 3	tier - 3	tier - 3	R1013	
1998	Jul	27	0	637.26	tier - 3	tier - 3	tier - 3	R1013	
2001	Nov	20	0	646.14	tier - 3	tier - 3	tier - 3	R1012	
1993	Jun	1	0	650	tier - 3	tier - 3	tier - 3	R1013	
1995	Jul	4	0	650	tier - 3	tier - 3	tier - 3	R1013	
2002	Nov	29	0	668	tier - 3	tier - 3	tier - 2	R1012	
1997	Nov	9	0	670	tier - 3	tier - 3	tier - 3	R1013	
2001	Sep	12	0	687.54	tier - 3	tier - 3	tier - 2	R1013	
1999	Dec	26	0	700	?	tier - 2	tier - 3	R1013	
1999	Dec	14	0	722.99	tier - 3	tier - 3	tier - 1	R1013	

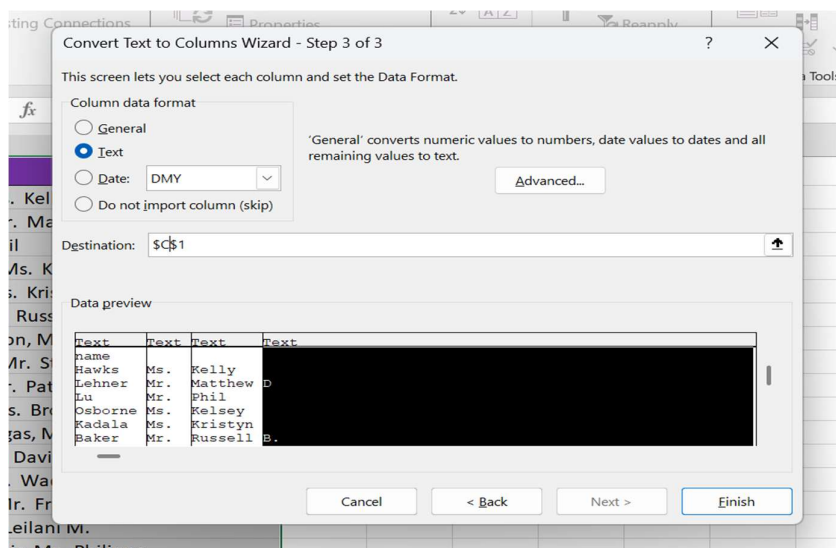
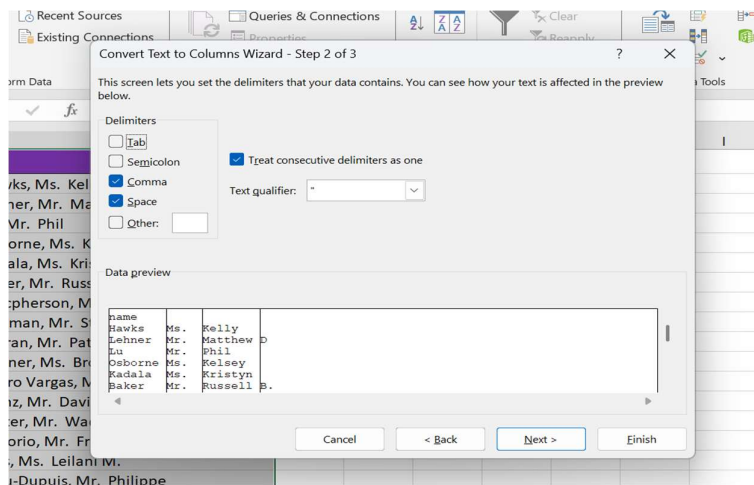
=IF(H19="?",INDEX(\$H\$2:\$H\$2344,MODE(MATCH(\$H\$2:\$H\$2344,\$H\$2:\$H\$2344,0))),H19)									
Year	Month	date	children	charge	Hospital tier	City tier	City tier	State	
1996	Sep	18	0	770.38	tier - 3	?	tier - 2	R1012	

State Id: The missing State Ids are imputed as “Unknown” using the “IF” function based on the given instructions.

=IF(I546="?", "Unknown", I546)							
date	children	charge	Hospital tier	City tier	State	State	
1	3	4827.9	tier - 1	tier - 2	?	Unknown	
5	1	37165.16	tier - 1	tier - 3	?	Unknown	

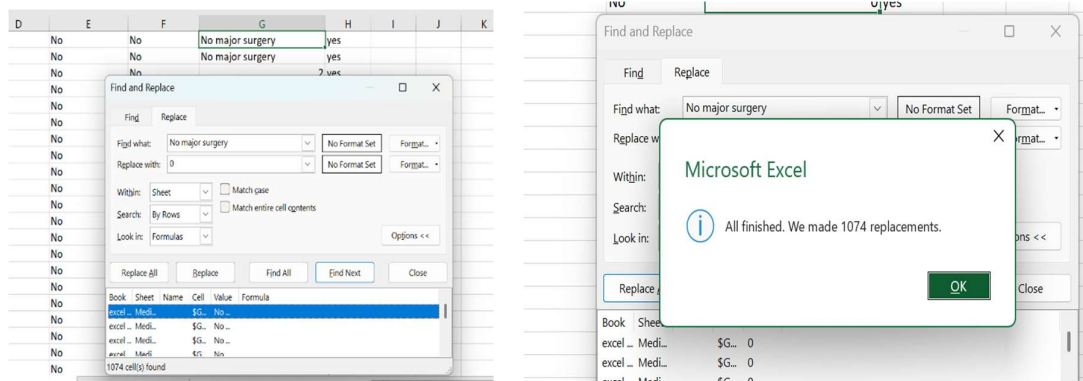
Data Transformation:

The ‘Names’ column in the “Customer Names” table is split into 3 meaningful columns: ‘Title’, ‘First Name’, and ‘Last Name’ using the function “Text to column”.



	A	B	C	D	E
1	Customer ID	Title	First Name	Last Name	
2	Id1	Hawks	Kelly		
3	Id2	Lehner	Matthew	D	
4	Id3	Lu	Phil		
5	Id4	Osborne	Kelsey		
6	Id5	Kadala	Kristyn		
7	Id6	Baker	Russell	B.	
8	Id7	Macpherson	Scott		
9	Id8	Hallman	Stephen		
10	Id9	Moran	Patrick	R.	
11	Id10	Benner	Brooke	N.	
12	Id11	Fierro Vargas	Paola	Andrea	
13	Id12	Franz	David		
14	Id13	Foster	Wade		
15	Id14	Tenorio	Franklin		
16	Id15	Rios	Leilani	M.	
17	Id16	Viau-Dupuis	Philippe		
18	Id17	Cronin	Jennifer	A.	
19	Id18	Noordstar	Christina	M.	

The text “no major surgery” in the column "Number of Major Surgeries" in "Medical Examinations" table is replaced by the numerical values 0 using “Find and Replace”



The new column named “Weight Status” was created using given below table and “IFS” function based on BMI values.

BMI	Weight Status
Below 18.5	Underweight
18.5 – 24.9	Normal Weight
25.0 – 29.9	Overweight
30.0 and Above	Obesity

customer ID	BMI	Weight Status	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	Smoking
1	47.41	Obesity	7.47	No	No	No	0	yes
12	30.36	Obesity	5.77	No	No	No	0	yes
13	34.485	Obesity	11.87	yes	No	No	2	yes
14	38.095	Obesity	6.05	No	No	No	0	yes
15	35.53	Obesity	5.45	No	No	No	0	yes
16	32.8	Obesity	6.59	No	No	No	0	yes
17	36.4	Obesity	6.07	No	No	No	0	yes
18	36.96	Obesity	7.93	No	No	No	3	yes
19	41.14	Obesity	9.58	yes	No	Yes	1	yes
110	38.06	Obesity	10.79	No	No	No	0	yes
111	37.7	Obesity	5.96	yes	No	No	2	yes
112	42.13	Obesity	11.9	No	No	No	0	yes
113	40.92	Obesity	8.41	No	No	No	0	yes
114	40.565	Obesity	7.02	No	No	No	0	yes
115	36.385	Obesity	7.59	yes	No	No	2	yes
116	39.9	Obesity	11.32	No	No	No	0	yes

The new column ‘Date of Birth’ was created by merged the ‘year’, ‘month’ and ‘date’ columns in the “Hospitalization Details” table using the formula **=DATEVALUE(CONCATENATE(D2, “”,C2, “”,B2))**. The custom format ‘DD-MMM-YYYY’ is used in this column.

customer Id	Year	Month	date	Date of Birth	children	charges	Hospital tier	City tier	State ID
2335	1992	Jul	9	09-Jul-1992	0	563.84	tier - 2	tier - 3	R1013
2334	1992	Nov	30	30-Nov-1992	0	570.62	tier - 2	tier - 1	R1013
2333	1993	Jun	30	30-Jun-1993	0	600	tier - 2	tier - 1	R1013
2332	1992	Sep	13	13-Sep-1992	0	604.54	tier - 3	tier - 3	R1013
2331	1998	Jul	27	27-Jul-1998	0	637.26	tier - 3	tier - 3	R1013
2330	2001	Nov	20	20-Nov-2001	0	646.14	tier - 3	tier - 3	R1012
2329	1993	Jun	1	01-Jun-1993	0	650	tier - 3	tier - 3	R1013
2328	1995	Jul	4	04-Jul-1995	0	650	tier - 3	tier - 3	R1013
2327	2002	Nov	29	29-Nov-2002	0	668	tier - 3	tier - 2	R1012
2326	1997	Nov	9	09-Nov-1997	0	670	tier - 3	tier - 3	R1013
2325	2001	Sep	12	12-Sep-2001	0	687.54	tier - 3	tier - 2	R1013
2324	1999	Dec	26	26-Dec-1999	0	700	tier - 2	tier - 3	R1013
2323	1999	Dec	14	14-Dec-1999	0	722.99	tier - 3	tier - 1	R1013
2322	2002	Sep	19	19-Sep-2002	0	750	tier - 3	tier - 1	R1012
2321	1993	Aug	9	09-Aug-1993	0	760	tier - 3	tier - 1	R1013
2320	1996	Oct	22	22-Oct-1996	0	760	tier - 3	tier - 3	R1013
2319	1993	Jun	28	28-Jun-1993	0	770	tier - 3	tier - 3	R1013

The ‘Age’ of each customer based on their ‘Date of Birth’ and the date of collection of the dataset, which is 8th June 2023 was calculated by using the formula “DATEDIF” and “Y”

=DATEDIF(B2,\$K\$3,"Y")										
Customer ID	Date of Birth	Age	children	charges	Hospital tier	City tier	State ID			
5	09-Jul-1992	30	0	\$563.84	tier - 2	tier - 3	R1013			
4	30-Nov-1992	30	0	\$570.62	tier - 2	tier - 1	R1013			08-Jun-23
3	30-Jun-1993	29	0	\$600.00	tier - 2	tier - 1	R1013			
2	13-Sep-1992	30	0	\$604.54	tier - 3	tier - 3	R1013			
1	27-Jul-1998	24	0	\$637.26	tier - 3	tier - 3	R1013			
0	20-Nov-2001	21	0	\$646.14	tier - 3	tier - 3	R1012			
9	01-Jun-1993	30	0	\$650.00	tier - 3	tier - 3	R1013			
8	04-Jul-1995	27	0	\$650.00	tier - 3	tier - 3	R1013			
7	29-Nov-2002	20	0	\$668.00	tier - 3	tier - 2	R1012			
6	09-Nov-1997	25	0	\$670.00	tier - 3	tier - 3	R1013			
5	12-Sep-2001	21	0	\$687.54	tier - 3	tier - 2	R1013			
4	26-Dec-1999	23	0	\$700.00	tier - 2	tier - 3	R1013			
3	14-Dec-1999	23	0	\$722.99	tier - 3	tier - 1	R1013			
2	19-Sep-2002	20	0	\$750.00	tier - 3	tier - 1	R1012			
1	09-Aug-1993	29	0	\$760.00	tier - 3	tier - 1	R1013			
0	22-Oct-1996	26	0	\$760.00	tier - 3	tier - 3	R1013			
9	28-Jun-1993	29	0	\$770.00	tier - 3	tier - 3	R1013			
8	18-Sep-1996	26	0	\$773.38	tier - 3	tier - 2	R1012			
7	07-Dec-1995	27	0	\$773.54	tier - 3	tier - 2	R1013			
6	07-Oct-2004	18	0	\$830.57	tier - 3	tier - 2	R1011			

The ‘Charges’ column was formatted into currency (\$).

Age	children	charges	Hospital tier	City tier	State ID
30	0	\$563.84	tier - 2	tier - 3	R1013
30	0	\$570.62	tier - 2	tier - 1	R1013
29	0	\$600.00	tier - 2	tier - 1	R1013
30	0	\$604.54	tier - 3	tier - 3	R1013
24	0	\$637.26	tier - 3	tier - 3	R1013
21	0	\$646.14	tier - 3	tier - 3	R1012
30	0	\$650.00	tier - 3	tier - 3	R1013
27	0	\$650.00	tier - 3	tier - 3	R1013
20	0	\$668.00	tier - 3	tier - 2	R1012
25	0	\$670.00	tier - 3	tier - 3	R1013
21	0	\$687.54	tier - 3	tier - 2	R1013
23	0	\$700.00	tier - 2	tier - 3	R1013
23	0	\$722.99	tier - 3	tier - 1	R1013
20	0	\$750.00	tier - 3	tier - 1	R1012

AI-Driven Data Exploration, Analysis & Visualization:

Data Preparation:

A new sheet named “**Healthcare**” was created using the above three table using the common column “Customer ID”. With the help of XLOOKUP formula the healthcare sheet was created. It contains the following columns, Customer ID, First Name, BMI, HBA1C, Heart Issues, Any Transplants, Cancer History, Number of Major Surgeries, Smoker, Weight Status, Diabetes Status, Date of Birth, Charges, Hospital Tier, City Tier, State ID, and Age. The diabetic status was filled with “**IFS**” function based on the “HBA1C” range given below.

HBA1C	Diabetic status
Normal	Below 5.7
prediabetes	5.7-6.4
Diabetes	Above 6.5

Get & Transform Data

Queries & Connections

Sort & Filter

Data tools

B2



=XLOOKUP(A2,'Customer Names'!\$A\$2:\$A\$2336,'Customer Names'!\$C\$2:\$C\$2336,"N/A")

	A	B	C	D	E	F	G	H	I
1	Customer Id	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
2	Id1	Kelly							
3	Id2	Matthew							
4	Id3	Phil							
5	Id4	Kelsey							
6	Id5	Kristyn							
7	Id6	Russell							
8	Id7	Scott							
9	Id8	Stephen							
10	Id9	Patrick							

Get & Transform Data

Queries & Connections

Sort & Filter

Data Tools

2

<

ata

From Table/Range

All

Workbook Links

Advanced

Columns

Get & Transform Data

Queries & Connections

Sort & Filter

Data Tools

2

Get & Transform Data

Queries & Connections

Sort & Filter

Columns

Data Tools

2

Get & Transform Data		Queries & Connections		Sort & Filter		Data Tools	
2		=XLOOKUP(A2,'Medical Examinations'!\$A\$2:\$A\$2336,'Medical Examinations'!\$F\$2:\$F\$2336,"N/A")					
B	C	D	E	F	G	H	I
First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
Kelly	47.41	7.47	No	No			
Matthew	30.36	5.77	No	No			
Phil	34.485	11.87	yes	No			
Kelsey	38.095	6.05	No	No			
Kristyn	35.53	5.45	No	No			
Russell	32.8	6.59	No	No			
Scott	36.4	6.07	No	No			
Stephen	36.96	7.93	No	No			
Patrick	41.14	9.58	yes	No			

Get & Transform Data		Queries & Connections		Sort & Filter		Data Tools		
=XLOOKUP(A2,'Medical Examinations'!\$A\$2:\$A\$2336,'Medical Examinations'!\$G\$2:\$G\$2336,"N/A")								
B	C	D	E	F	G	H	I	
er Id	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker
	Kelly	47.41	7.47	No	No	No		
	Matthew	30.36	5.77	No	No	No		
	Phil	34.485	11.87	yes	No	No		
	Kelsey	38.095	6.05	No	No	No		
	Kristyn	35.53	5.45	No	No	No		
	Russell	32.8	6.59	No	No	No		
	Scott	36.4	6.07	No	No	No		
Get & Transform Data		Queries & Connections		Sort & Filter		Data Tools		
=XLOOKUP(A2,'Medical Examinations'!\$A\$2:\$A\$2336,'Medical Examinations'!\$H\$2:\$H\$2336,"N/A")								
C	D	E	F	G	H	I		
me	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	
	47.41	7.47	No	No	No	0		
	30.36	5.77	No	No	No	0		
	34.485	11.87	yes	No	No	2		
	38.095	6.05	No	No	No	0		
	35.53	5.45	No	No	No	0		
	32.8	6.59	No	No	No	0		
	36.4	6.07	No	No	No	0		
	36.96	7.93	No	No	No	3		
	41.14	9.58	yes	No	Yes	1		
	38.06	10.79	No	No	No	0		
	37.7	5.96	yes	No	No	2		
Get & Transform Data		Queries & Connections		Sort & Filter		Data Tools		
=XLOOKUP(A2,'Medical Examinations'!\$A\$2:\$A\$2336,'Medical Examinations'!\$I\$2:\$I\$2336,"N/A")								
B	C	D	E	F	G	H	I	
First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	
Kelly	47.41	7.47	No	No	No		yes	
Matthew	30.36	5.77	No	No	No		yes	
Phil	34.485	11.87	yes	No	No		yes	
Kelsey	38.095	6.05	No	No	No		yes	
Kristyn	35.53	5.45	No	No	No		yes	
Russell	32.8	6.59	No	No	No		yes	
Scott	36.4	6.07	No	No	No		yes	
Stephen	36.96	7.93	No	No	No		yes	
Patrick	41.14	9.58	yes	No	Yes		yes	
Brooke	38.06	10.79	No	No	No		yes	

File/Range

Refresh

Workbook Links

Sort

Filter

Advanced

Text to Columns

Data Tools

Window

Forecast

Get & Transform Data

Queries & Connections

Sort & Filter

Data Tools

Forecast

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fx

=XLOOKUP(A2,'Medical Examinations'!\$A\$2:\$A\$2336,'Medical Examinations'!\$C\$2:\$C\$2336,"N/A")

B	C	D	E	F	G	H	I	J
First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status
Kelly	47.41	7.47	No	No	No		0 yes	Obesity
Matthew	30.36	5.77	No	No	No		0 yes	Obesity
Phil	34.485	11.87	yes	No	No		2 yes	Obesity
Kelsey	38.095	6.05	No	No	No		0 yes	Obesity
Kristyn	35.53	5.45	No	No	No		0 yes	Obesity
Russell	32.8	6.59	No	No	No		0 yes	Obesity
Scott	36.4	6.07	No	No	No		0 yes	Obesity
Stephen	36.96	7.93	No	No	No		3 yes	Obesity
Patrick	41.14	9.58	yes	No	Yes		1 yes	Obesity
Brooke	38.06	10.79	No	No	No		0 yes	Obesity

fx

=IFS(D2>=6.5,"Diabetes",D2>=5.7,"Prediabetes",D2<5.7,"Normal",D2="N/A",D2)

C	D	E	F	G	H	I	J	K	L
BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth
47.41	7.47	No	No	No		0 yes	Obesity	Diabetes	
30.36	5.77	No	No	No		0 yes	Obesity	Prediabetes	
34.485	11.87	yes	No	No		2 yes	Obesity	Diabetes	
38.095	6.05	No	No	No		0 yes	Obesity	Prediabetes	
35.53	5.45	No	No	No		0 yes	Obesity	Normal	
32.8	6.59	No	No	No		0 yes	Obesity	Diabetes	
36.4	6.07	No	No	No		0 yes	Obesity	Prediabetes	
36.96	7.93	No	No	No		3 yes	Obesity	Diabetes	
41.14	9.58	yes	No	Yes		1 yes	Obesity	Diabetes	
38.06	10.79	No	No	No		0 yes	Obesity	Diabetes	
37.7	5.96	yes	No	No		2 yes	Obesity	Prediabetes	

fx

=XLOOKUP(A2,'Hospitalisation Details'!\$A\$2:\$A\$2341,'Hospitalisation Details'!\$B\$2:\$B\$2341)

H	I	J	K	L	M	N
Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hosp
	0 yes	Obesity	Diabetes	04-Aug-1962		
	0 yes	Obesity	Prediabetes	27-Oct-1994		
	2 yes	Obesity	Diabetes	27-Jun-1958		
	0 yes	Obesity	Prediabetes	04-Sep-1963		
	0 yes	Obesity	Normal	29-Dec-1978		
	0 yes	Obesity	Diabetes	22-Jul-1959		
	0 yes	Obesity	Prediabetes	27-Oct-1965		
	3 yes	Obesity	Diabetes	11-Oct-1962		
	1 yes	Obesity	Diabetes	01-Dec-1968		
	0 yes	Obesity	Diabetes	21-Dec-1961		
	2 yes	Obesity	Prediabetes	27-Aug-1962		
	0 yes	Obesity	Diabetes	16-Nov-1958		

Alignment		Number		Styles				
=XLOOKUP(A2,'Hospitalisation Details'!\$A\$2:\$A\$2341,'Hospitalisation Details'!\$E\$2:\$E\$2341)								
I	J	K	L	M	N	O	P	Q
Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital T	City Tier	State ID	Age
Yes	Obesity	Diabetes	04-Aug-1962	\$52,590.83				
Yes	Obesity	Prediabetes	27-Oct-1994	\$51,194.56				
Yes	Obesity	Diabetes	27-Jun-1958	\$49,577.66				
Yes	Obesity	Prediabetes	04-Sep-1963	\$48,970.25				
Yes	Obesity	Normal	29-Dec-1978	\$48,885.14				
Yes	Obesity	Diabetes	22-Jul-1959	\$48,824.45				
Yes	Obesity	Prediabetes	27-Oct-1965	\$48,675.52				
Yes	Obesity	Diabetes	11-Oct-1962	\$48,673.56				
Yes	Obesity	Diabetes	01-Dec-1968	\$48,549.18				
Yes	Obesity	Diabetes	21-Dec-1961	\$48,517.56				

Home														Insert														Draw														Page Layout														Formulas														Data														Review														View														Developer														Help														Tell me what you want to 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	F	G	H	I	J	K	L	M	N	O	P
es	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID
	No	No		0 yes	Obesity	Diabetes	12-Oct-1968	\$63,770.43	tier - 1	tier - 3	
	No	No		0 yes	Obesity	Prediabetes	08-Jun-1977	\$62,592.87	tier - 2	tier - 3	
	No	No		2 yes	Obesity	Diabetes	11-Sep-1970	\$60,021.40	tier - 1	tier - 1	
	No	No		0 yes	Obesity	Prediabetes	06-Jun-1991	\$58,571.07	tier - 1	tier - 3	
	No	No		0 yes	Obesity	Normal	19-Jun-1989	\$55,135.40	tier - 1	tier - 2	
	No	No		0 yes	Obesity	Diabetes	04-Aug-1962	\$52,590.83	tier - 1	tier - 3	
	No	No		0 yes	Obesity	Prediabetes	27-Oct-1994	\$51,194.56	tier - 1	tier - 3	
	No	No		3 yes	Obesity	Diabetes	27-Jun-1958	\$49,577.66	tier - 2	tier - 2	
	No	Yes		1 yes	Obesity	Diabetes	04-Sep-1963	\$48,970.25	tier - 1	tier - 2	
	No	No		0 yes	Obesity	Diabetes	29-Dec-1978	\$48,885.14	tier - 1	tier - 2	
	No	No		2 yes	Obesity	Prediabetes	22-Jul-1959	\$48,824.45	tier - 2	tier - 1	
	No	No		0 yes	Obesity	Diabetes	27-Oct-1965	\$48,675.52	tier - 1	tier - 2	
	No	No		0 yes	Obesity	Diabetes	11-Oct-1962	\$48,673.56	tier - 1	tier - 2	
	No	No		0 yes	Obesity	Diabetes	01-Dec-1968	\$48,549.18	tier - 1	tier - 3	
	No	No		2 yes	Obesity	Diabetes	21-Dec-1961	\$48,517.56	tier - 1	tier - 3	
	No	No		0 yes	Obesity	Diabetes	27-Aug-1962	\$48,173.36	tier - 1	tier - 3	
	No	No		3 yes	Obesity	Diabetes	16-Nov-1958	\$47,928.03	tier - 2	tier - 3	

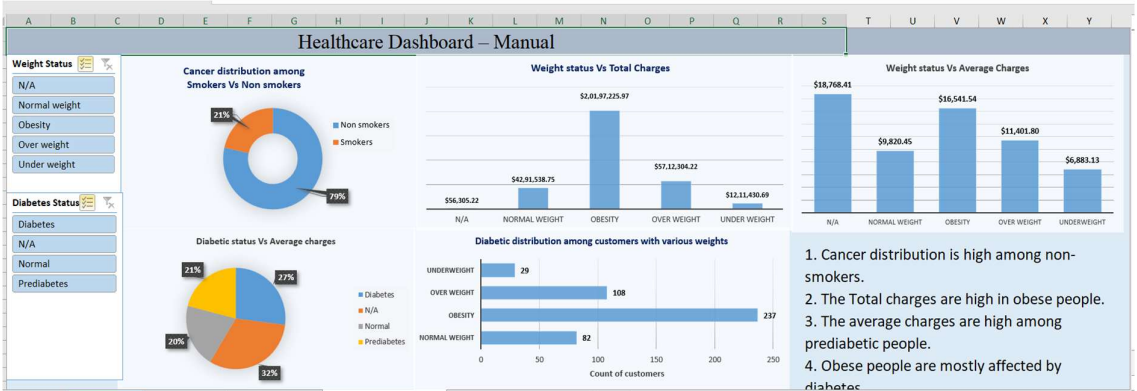
	F	G	H	I	J	K	L	M	N	O	P	Q
ues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID	Age
	No	No		0 yes	Obesity	Diabetes	12-Oct-1968	\$63,770.43	tier - 1	tier - 3	R1013	
	No	No		0 yes	Obesity	Prediabetes	08-Jun-1977	\$62,592.87	tier - 2	tier - 3	R1013	
	No	No		2 yes	Obesity	Diabetes	11-Sep-1970	\$60,021.40	tier - 1	tier - 1	R1012	
	No	No		0 yes	Obesity	Prediabetes	06-Jun-1991	\$58,571.07	tier - 1	tier - 3	R1024	
	No	No		0 yes	Obesity	Normal	19-Jun-1989	\$55,135.40	tier - 1	tier - 2	R1012	
	No	No		0 yes	Obesity	Diabetes	04-Aug-1962	\$52,590.83	tier - 1	tier - 3	R1011	
	No	No		0 yes	Obesity	Prediabetes	27-Oct-1994	\$51,194.56	tier - 1	tier - 3	R1011	
	No	No		3 yes	Obesity	Diabetes	27-Jun-1958	\$49,577.66	tier - 2	tier - 2	R1013	
	No	Yes		1 yes	Obesity	Diabetes	04-Sep-1963	\$48,970.25	tier - 1	tier - 2	R1013	
	No	No		0 yes	Obesity	Diabetes	29-Dec-1978	\$48,885.14	tier - 1	tier - 2	R1013	
	No	No		2 yes	Obesity	Prediabetes	22-Jul-1959	\$48,824.45	tier - 2	tier - 1	R1011	
	No	No		0 yes	Obesity	Diabetes	27-Oct-1965	\$48,675.52	tier - 1	tier - 2	R1013	
	No	No		0 yes	Obesity	Diabetes	11-Oct-1962	\$48,673.56	tier - 1	tier - 2	R1013	
	No	No		0 yes	Obesity	Diabetes	01-Dec-1968	\$48,549.18	tier - 1	tier - 3	R1016	
	No	No		2 yes	Obesity	Diabetes	21-Dec-1961	\$48,517.56	tier - 1	tier - 3	R1024	
	No	No		0 yes	Obesity	Diabetes	27-Aug-1962	\$48,173.36	tier - 1	tier - 3	R1011	

	F	G	H	I	J	K	L	M	N	O	P	Q
ues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID	Age
	No	No		0 yes	Obesity	Diabetes	12-Oct-1968	\$63,770.43	tier - 1	tier - 3	R1013	54
	No	No		0 yes	Obesity	Prediabetes	08-Jun-1977	\$62,592.87	tier - 2	tier - 3	R1013	46
	No	No		2 yes	Obesity	Diabetes	11-Sep-1970	\$60,021.40	tier - 1	tier - 1	R1012	52
	No	No		0 yes	Obesity	Prediabetes	06-Jun-1991	\$58,571.07	tier - 1	tier - 3	R1024	32
	No	No		0 yes	Obesity	Normal	19-Jun-1989	\$55,135.40	tier - 1	tier - 2	R1012	33
	No	No		0 yes	Obesity	Diabetes	04-Aug-1962	\$52,590.83	tier - 1	tier - 3	R1011	60
	No	No		0 yes	Obesity	Prediabetes	27-Oct-1994	\$51,194.56	tier - 1	tier - 3	R1011	28
	No	No		3 yes	Obesity	Diabetes	27-Jun-1958	\$49,577.66	tier - 2	tier - 2	R1013	64
	No	Yes		1 yes	Obesity	Diabetes	04-Sep-1963	\$48,970.25	tier - 1	tier - 2	R1013	59
	No	No		0 yes	Obesity	Diabetes	29-Dec-1978	\$48,885.14	tier - 1	tier - 2	R1013	44
	No	No		2 yes	Obesity	Prediabetes	22-Jul-1959	\$48,824.45	tier - 2	tier - 1	R1011	63
	No	No		0 yes	Obesity	Diabetes	27-Oct-1965	\$48,675.52	tier - 1	tier - 2	R1013	57
	No	No		0 yes	Obesity	Diabetes	11-Oct-1962	\$48,673.56	tier - 1	tier - 2	R1013	60
	No	No		0 yes	Obesity	Diabetes	01-Dec-1968	\$48,549.18	tier - 1	tier - 3	R1016	54
	No	No		2 yes	Obesity	Diabetes	21-Dec-1961	\$48,517.56	tier - 1	tier - 3	R1024	61
	No	No		0 yes	Obesity	Diabetes	27-Aug-1962	\$48,173.36	tier - 1	tier - 3	R1011	60

Manual Dashboard

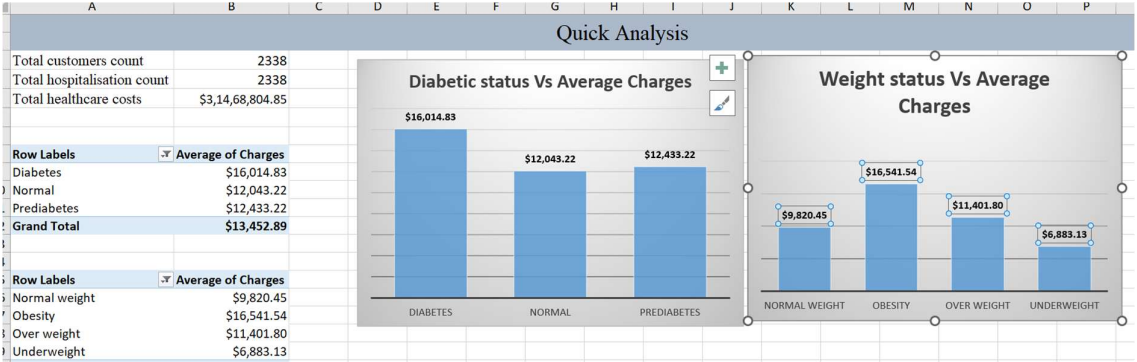
Pivot tables are created manually using the combined healthcare dataset. A donut chart was created based on cancer distribution among smokers Vs non-smokers. The clustered column chart was created based on charges by weight status. Diabetic distribution among the customers

based on various weight and Weight status Vs Average Charges charts are also created. The “Dashboard-Manual” was created in separate sheet and insights are marked in the same sheet.



AI-Powered Summary Statistics & Visualization

The new sheet “Healthcare” was highlighted from other sheets.Using “Data Analysis” tool in the Data tab the automatically generated statistical analysis for healthcare data was created in separate sheet. When we select the healthcare column, the “Quick Analysis” tool was appeared in the sheet. Using the “Quick analysis” total customers count, total hospitalisation count, total healthcare costs, average costs and medical metrics are calculated and marked in separate sheet **Quick Analysis**. The recommended charts also added in this sheet.



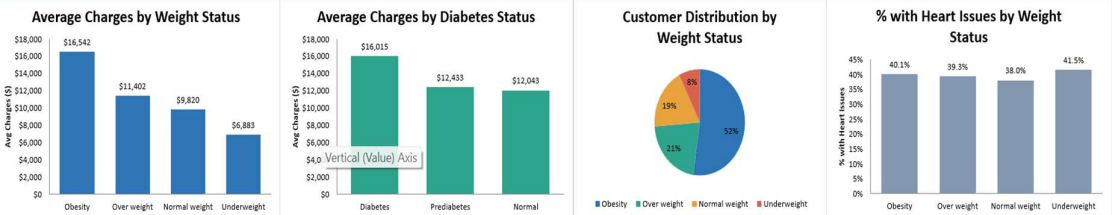
AI-Dashboard Creation

Using “Claude AI” the “Healthcare Dashboard – AI Analysis” was created in separate sheet.

Healthcare Dashboard – AI Analysis

Filters Weight Status: All Diabetes Status: All Click cells above and choose a value — every chart & KPI updates instantly

Total Customers	Average Charges	Average BMI	Average HBA1C	% Smokers	% Diabetic
2338	\$13,460	31.0	6.58	20.9%	33.5%



Customer Distribution by Diabetes Status	Average Charges: Smoker vs Non-Smoker	Average Charges by Hospital Tier	Average BMI by Diabetes Status
Status	Smoker	Tier	